

Samreddy Shashanth Reddy

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PROFESSIONAL SUMMARY

Results-driven Data Professional with expertise in Data Science, Machine Learning Engineering, Data Analytics, and Data Engineering. Skilled in building and deploying scalable ML systems, developing data pipelines, and deriving actionable insights through EDA. Proficient in Python, SQL, TensorFlow, cloud platforms, MLOps, and NLP technologies such as BERT and Transformers.

EDUCATION

M.S. Data Science	Sep 2023 - May 2025
New Jersey Institute of Technology	New Jersey
B.S. Computer Science	Jul 2019 - May 2023
Vellore Institute of Technology	Vellore

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, JavaScript, R, SQL, MySQL, MATLAB

Data Engineering & Cloud: ETL Pipelines, Apache Kafka, Airflow, AWS S3, Redshift, Azure Data Factory, GCP BigQuery ,DataBricks, Data Collection Systems, Statistical Modeling, Secondary Data Sources

Advanced SQL: Window Functions, CTEs, Query Optimization ,MS SQL Server

MLOps& Deployment: Docker, Kubernetes, Model Deployment using Flask/FastAPI

Machine Learning & AI: NumPy, Pandas, scikit-learn, TensorFlow, PyTorch ,Keras, SciPy, Feature Engineering, Hyperparameter Tuning

Deep Learning & NLP: LSTM, BERT, Transformers, Sentiment Analysis, Text Classification

A/B Testing & Experimentation: Statistical Testing, Business Impact Analysis

Big Data & Cloud Platforms: Apache Hadoop, Spark, AWS, Azure, NoSQL

Data Visualization & BI Tools: Tableau, Power BI, Looker, Google Analytics, Excel

Data Analysis & Statistical Methods: Analyzing Data, Data Mining, Data Manipulation, Quantitative Analysis, Statistical Methods, Emerging Patterns Complex Data Sets

Synthesize Results, Actionable Information, Analytic Models, Analytical SQL Queries, Estimation Software

Modeling & Data Processing Techniques: DataStage, Modeling Techniques, Data Visualization Techniques

Soft Skills: Storytelling with Data, Cross-functional Collaboration, Stakeholder Communication

Data Management & Governance: Data Assets ,Data Governance ,Data Management Principles ,Information Quality Control ,Data Quality Issues

Core Competencies: Drive Innovation , Build Data Models , Data Solutions , Analyze Data , Work with Multiple Sources , Data Pipelines , Exploratory Data Analysis , Data Science Techniques , Natural Language Processing , Machine Learning , Model Deployment

EXPERIENCE

Data Analyst Intern	Jan 2023 - Jul 2023
Defense Research & Development Org.	
- Boosted model performance by 25% via advanced preprocessing, feature extraction, and SQL-based data engineering. - Revamped anomaly detection algorithms using computer vision and deep learning, improving accuracy by 25%. - Architected efficient ETL pipelines, seamlessly integrating Power BI & Tableau dashboards to slash manual analysis time by 60% - Implemented a real-time detection system utilizing Autoencoders, Kafka, TensorFlow & Prometheus, boosting anomaly capture by 50% across 1M+ events/day, leveraging the Variational Autoencoder model - Optimized deployment time by 33% & facilitated rapid monitoring through Grafana by containerizing models with Docker & Kubernetes & leveraging Prometheus for monitoring	

PROJECTS

Multi-Modal Data Fusion for Product Classification

- Transformed over 50,000 product listings into a structured data model using a sophisticated multi-input deep learning architecture, blending CNNs for images & **BERT for text analysis**, resulting in a 20% accuracy boost & 30% faster processing
- Improved classification accuracy by 22% by integrating NLP and visual features, enhancing product tagging efficiency.
- Employed TensorFlow to enhance data pipelines for real-time classification, facilitating **profound analysis across diverse modalities like Images, Text, & Audio**, leading to a remarkable 70% surge in efficiency

Sales Forecasting

- Deployed data science techniques like ARIMA and LSTM to forecast sales, boosting business revenue projections with 20% improved accuracy.
- Conducted in-depth exploratory data analysis (EDA) on sales data with Tableau & Excel, **pinpointing trends & seasonality** to inform model selection
- Developed scalable models for real-time predictions with FastAPI, Docker, Kubernetes, & TensorFlow technologies achieving a 99% accuracy rate & 50% processing time reduction

Credit Card Fraud Detection

- Engineered fraud detection pipelines using **Autoencoders, Decision Trees, and XGBoost**, reducing false positives by **18%**.
- Built a real-time scoring engine that ingested transaction data from **Kafka topics**, classifying fraud on-the-fly with **<200ms latency**.
- Enhanced model performance by 40% through the integration of Random Forest & Logistic Regression models & employing **SMOTE & cross-validation techniques** to address class imbalance, fortify model resilience, & improve accuracy & precision metrics evaluated through F1 score & Area under ROC curve

Real-Time Anomaly Detection Pipeline

- Integrated streaming data sources with **Apache Kafka** and monitored performance using **Prometheus + Grafana**, improving anomaly response time by **40%**.
- Built and deployed autoencoder-based models to detect anomalies in sensor data, trained on **1M+ event logs**.
- Created reusable **Dockerized workflows** with **CI/CD pipelines**, reducing deployment time by **33%**.
- Elevated productivity by 47% by implementing comprehensive **Grafana dashboards & Kibana log visualizations**, revealing key performance trends

Deep Learning-Based Sentiment Analysis

- Applied **BERT fine-tuning** on IMDB-style sentiment data, achieving **96% classification accuracy** on 50K+ customer reviews.
- Designed an interactive Tableau dashboard to visualize sentiment trends within diverse product categories, leading to a **20% increase in engagement** while effectively pinpointing key sentiment drivers across Clothing, Electronics, & Home Goods
- Constructed a real-time API using **Flask**, delivering sentiment scores instantly for customer review pages.

CERTIFICATIONS

- Google Data Analytics Professional Certificate
- IBM Data Analyst Professional Certificate
- Tableau Certified Data Analyst
- Microsoft Azure Fundamentals

LEADERSHIP & VOLUNTEERING

- Organization: Blockchain Workshop
- As an Event Coordinator led a 100+ attendee event, collaborating with IIT experts to deliver hands-on sessions.
- Organization: Adventure Club
- Spearheaded 7 university-wide competitions and NGO campaigns focused on social impact, boosting student participation by 40%